Shiva Chethan Halamane

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	8.81
Narayana Olympiad School Sahakaranagar Banglore KK(45673)	Class XII (CBSE)	2024	96.60%
Vivekananda English Medium School Tenkila Puttur	Class X (KSEEB (Karnataka State Board))	2022	99.68%

SCHOLASTIC ACHIEVEMENTS

- Achieved the title of Expert on Codeforces with a peak rating of 1646 . (Handle : Adamant18)
- Achieved Rank 179 in the preliminary round of ICPC (International Collegiate Programming Contest) .
- Secured Rank 687 in IICPC QuantFest among 10,000+ participants.
- Amongst the center top 10% performers in NSEC, NSEP , NSEA .
- Secured International Rank 7 in the International Mathematics Olympiad by the Science Olympiad Foundation .

PROJECTS

RISC-V Multi-Stage Pipelined Simulator | Computer Architecture (C++)

- Implemented a **5-stage pipelined RISC-V simulator** with Instruction Fetch, Decode, Execute, Memory Access, and Write Back stages.
- Implemented a **hazard detection unit** with **pipeline stalling** and **data forwarding** to resolve data hazards and ensure correct execution.
- Implemented **Single-Bit** and **N-Bit branch predictors** to improve branch prediction accuracy and reduce control hazards by leveraging branch history.
- Developed an **L1 byte-addressable cache** simulator supporting **LRU, FIFO, and Random** replacement policies.

xv6 Operating System | Operating Systems-II (C language)

- Extended the baseline xv6 operating system with advanced OS-level features and performance enhancements.
- Implemented a **4-level Multi-Level Feedback Queue (MLFQ)** scheduler to improve process management, reduce starvation, and ensure fair CPU allocation.
- Implemented disk-backed swapping using the **clock page replacement algorithm**, along with page fault handling for efficient virtual memory management.
- Integrated **RAID 0 , RAID 1 , and RAID 5** to simulate fault tolerance, striping, and parity-based data recovery, with **FCFS/SSTF disk scheduling**.

Infinite Precision Arithmetic Calculator | Software Development Fundamentals (Java)

- Developed custom **AInteger** and **AFloat** classes to support arbitrary-precision integer and floating-point arithmetic beyond existing Java data types.
- Implemented **signed arbitrary-precision arithmetic**, including addition, subtraction, multiplication, and division for arbitrarily large integers and high-precision floating-point numbers, with efficient sign handling and normalization.
- Developed algorithms for **high-precision decimal operations**, including decimal alignment, normalization, and user-defined precision handling.

Sudoku Verifier | (C++)

- Implemented a multithreaded Sudoku verifier in C++ using Pthreads, with mutex locks for concurrency control.
- Parallelized validation of Sudoku rows, columns, and subgrids to utilize multicore processors and reduce verification time.
- Implemented thread-safe error reporting and output generation to ensure correct results during concurrent execution.

SKILLS

- **Programming Languages:** C, C++, RISC-V Assembly, Python, Java, SQL (MySQL, PostgreSQL), OCaml
- **Web Development:** HTML, CSS, JavaScript
- **Tools & Technologies:** Linux, Git, GitHub, LaTeX

RELEVANT COURSES

- **Mathematics:** Discrete Mathematics for Computer Science, Calculus, Probability and Random Variables, Applied Statistics.
- **Computer Science:** Introduction to Programming, Software Development Fundamentals, Data Structures, Algorithms, Operating Systems, Computer Architecture , Database Management Systems , Theory of Computation,

POSITIONS OF RESPONSIBILITY

- Appointed as a Senior Core Member of Infero, the Algorithmic and Logical Reasoning Club at IIT Hyderabad
- Worked as a Core Member of Infero, mentoring juniors in competitive programming and contributing to problem setting for Code Clash at Elan & nVision.
- Nominated for the role of Chess Coordinator at IIT Hyderabad .
- Served as Coordinator, Competitions Domain, Elan & nVision, IIT Hyderabad .
- Appointed as Events Coordinator, Diesta, IIT Hyderabad .

EXTRACURRICULAR

- FIDE-rated chess player with ratings of 1718 (Classical) and 1757 (Blitz).
- Qualified for the SGFI National Chess Championship for three consecutive years (2018-2020), including securing 1st place in the Karnataka State Selection (2018).
- Awarded the Purple Belt in Shotokan-style Karate.